

Concrete Classes

Cadastre

Grid Hosting Capacity Document assembly model

A singular complete record of all land parcels within some hierarchical level of administrative jurisdiction. The cadastre provides the ability to find the results of land parcel surveys typically used for land management and planning.

Native Members (European)

energyIdentCodeEic (European)	0..1	String	The attribute is used for an exchange of the EIC code (Energy identification Code). The length of the string is 16 characters as defined by the EIC code. For details on EIC scheme please refer to ENTSO-E web site.
shortName (European)	0..1	String	The attribute is used for an exchange of a human readable short name with length of the string 12 characters maximum.

Native Members

administrationLevel	0..1	String	Administrative level is referring to the organisation that is administrative authority for managing the a hierarchical division of a geographic area into land, property groups. This typically established by government that has the responsibility of managing governance, services, taxation, and legal jurisdiction, such as municipalities, townships, counties, or regions.
aliasName	0..1	String	The aliasName is free text human readable name of the object alternative to IdentifiedObject.name. It may be non unique and may not correlate to a naming hierarchy. The attribute aliasName is retained because of backwards compatibility between CIM releases. It is however recommended to replace aliasName with the Name class as aliasName is planned for retirement at a future time.
cadastralReference	0..1	String	Local or national unique identification reference for the cadastral for land parcel, ensuring effective land management, ownership tracking, and legal compliance.
description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
identifier	0..1	UUID	A universally unique object identifier. Used to uniquely identify persistent objects between CIM messages.
levelFourDivision	0..1	String	Level Four Division refers to a further subdivision of a Level Three Division within a cadastral hierarchy, typically used to represent specific units or sections within a shared property or structure, such as

			apartments, condominiums, or co-owned facilities.
levelOneDivision	0..1	String	Level One Division refers to the highest organizational or reference unit within a cadastral hierarchy, typically representing a primary division of land, such as a farm or property group, which can be further subdivided into smaller parcels.
levelThreeDivision	0..1	String	Level Three Division refers to a further subdivision of a Level Two Division within a cadastral hierarchy, typically used to represent specialized land units such as leased parcels, easements, or other secondary divisions associated with distinct legal or usage rights.
levelTwoDivision	0..1	String	Level Two Division refers to a subdivision of a Level One Parcel within a cadastral hierarchy, representing smaller, distinct land units typically used for individual ownership, usage, or development, such as specific plots or lots within a farm or property group.
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.

Circle

Grid Hosting Capacity Document assembly model

A circular area in space, described by a center point and radius. Any circle object must have a center and an outer radius and may have an inner radius. If an inner radius is specified, the area within the inner radius is to be excluded from the area within the outer radius, forming a ring shape.

Native Members

innerRadius	0..1	ESMP_Length	A circle may have an inner radius. If specified the area described by the inner radius is to be excluded from the area within the outer radius, forming a ring shape.
outerRadius	0..1	ESMP_Length	Every circle must have an outer radius which describes the outer limit of the circular area.
CircleCenter_LinearCoordinate	[0..1]	LinearCoordinate	A point representing the center of a circular area. Any circle object must have one CircleCenter.
CircleCenterPositionPoint	[0..1]	PositionPoint	A point representing the center of a circular area. Any circle object must have one CircleCenter.

EnvAsset

Grid Hosting Capacity Document assembly model

asset is an energy data space specific attribute referring to the Asset from which data is being requested

Native Members

meterId	0..1	String	meterId is an energy data space specific attribute that specifies the id of the meter that measures the asset.
operatorId	0..1	String	operatorId is an energy data space specific attribute that specifies the id operator of the asset referred to.
type	0..1	String	type is an energy data space specific attribute that specifies type of asset referred to.

Geometry

Grid Hosting Capacity Document assembly model

Geo-spatial representation of a physical location. A location may be represented as a Point, Line, Polygon, or Circle.

Native Members

coordinateSystem.crsUrn	0..1	String	A Uniform Resource Name (URN) for the coordinate reference system (crs) used to define 'Location.PositionPoints'. An example would be the European Petroleum Survey Group (EPSG) code for a coordinate reference system, defined in URN under the Open Geospatial Consortium (OGC) namespace as: urn:ogc:def:uom:EPSG::XXXX, where XXXX is an EPSG code (a full list of codes can be found at the EPSG Registry web site http://www.epsg-registry.org/). To define the coordinate system as being WGS84 (latitude, longitude) using an EPSG OGC, this attribute would be urn:ogc:def:uom:EPSG::4236. A profile should limit this code to a set of allowed URNs agreed to by all sending and receiving parties. --- The coordinate system used for a given geometry.
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.

GHC_Envelope

Grid Hosting Capacity Document assembly model

The envelope entity is a non-CIM object used in an energy data space environment to serve as a central top-level entity connecting the MarketDocumentHeader (energy data space specific context information block) with the regular CIM MarketDocument.

Native Members

MessageDocumentHeader	[0..1]	MessageDocumentHeader
GHC_MarketDocument	[0..1]	GHC_MarketDocument

GHC_MarketDocument

Grid Hosting Capacity Document assembly model

An electronic document containing the information necessary to satisfy the requirements of a given business process.

Native Members

createdDateTime	0..1	ESMP_DateTime	The date and time of the creation of the document.
description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
legalInformationText	0..1	String	Text with the required legal information from the information provider (Connecting System operator=CSO)
legalInformationWebsites	0..1	String	URL pointing to the required legal information from the information provider (Connecting System operator=CSO)
mRID	1..1	ID_String	The unique identification of the document being exchanged within a business process flow. In the ESMP context, the "model authority" is defined as a party (originator of the exchange) that provides an identification in the context of a business exchange such as document identification, ... Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
publishedDateTime	0..1	DateTime	The date time the legal information is published
receiver_MarketParticipant.marketRole.type	1..1	MarketRoleKind_String	The identification of the role played by a market player. --- The MarketParticipant associated with an electronic document header. --- The role associated with a

			MarketParticipant.
receiver_MarketParticipant.mRID	1..1	PartyID_String	The identification of a party in the energy market. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The MarketParticipant associated with an electronic document header.
receiver_MarketParticipant.name	0..1	String	The name is any free human readable and possibly non unique text naming the object. --- The MarketParticipant associated with an electronic document header.
revisionNumber	0..1	ESMPVersion_String	The identification of the version that distinguishes one evolution of a document from another.
sender_MarketParticipant.marketRole.type	1..1	MarketRoleKind_String	The identification of the role played by a market player. --- The MarketParticipant associated with an electronic document header. --- The role associated with a MarketParticipant.
sender_MarketParticipant.mRID	1..1	PartyID_String	The identification of a party in the energy market. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID

			is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The MarketParticipant associated with an electronic document header.
sender_MarketParticipant.name	0..1	String	The name is any free human readable and possibly non unique text naming the object. --- The MarketParticipant associated with an electronic document header.

HostingCapacity

Grid Hosting Capacity Document assembly model

A structure for a set of information on grid hosting capacity.

Native Members

capacityDataDetails	0..1	String	Informational field for further information on - timing (e.g. is time for capacity available for connection a snapshot or is it the time when connection can be performed) - description of the SO description of the data provided, - description of the meaning of the different colours
firstConnectionYear	0..1	Integer	The year that the capacity is applicable.
identifier	0..1	UUID	A universally unique object identifier. Used to uniquely identify persistent objects between CIM messages.
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.
publicationDate	0..1	Date	The date when the data was produced or when it can be available

HostingCapacityDetail

Grid Hosting Capacity Document assembly model

A structure for grid hosting capacity information at a certain year and duration for a certain customerconnectiontype.

Native Members

appliesFromYear	0..1	Integer	The year that the capacity is first applicable.
capacityAvailableForConnection	0..1	CEEDS_ActivePower	Information on firm capacity available for new connections in MW. Requirement for grid hosting capacity information in (EU) 2019/943 and (EU) 2019/944
capacityUnderConnectionRequest	0..1	CEEDS_ActivePower	Information on connections requests in MW. Requirement for grid hosting capacity information in (EU) 2019/943 and (EU) 2019/944
customerConnectionType	1..1	CustomerConnectionTypeKind	Information on customer connection types for which grid hosting capacity information is applicable. Possible values are: Consumption Production Energy storage Hybrid park
customerSubConnectionType	1..1	CustomerSubConnectionTypeKind	Information on customer connection types (subgroups) for which grid hosting capacity information is applicable. Possible values are: Solar Wind EV charger Industry
customerSubSubConnectionType	1..1	CustomerSubSubConnectionTypeKind	Information on customer connection types (subsubgroups) for which grid hosting capacity information is applicable. Possible values are: EV charger light vehicle EV charger heavy duty

			vehicle Industry - High usage Industry - Average usage
durationApplicable	0..1	Integer	The period in years starting from appliesYearFrom that the capacity is available. So: appliesYearFrom=2028 durationApplicable=2 resulting period: 2028-2030
FlexibleConnectionCapacity	0..1	CEEDS_ActivePower	Information on possibility of flexible connection capacity in MW. Requirement for grid hosting capacity information in (EU) 2019/943 and (EU) 2019/944
levelOfFlexibility	0..1	String	Description of flexible connection agreement (e.g. " 2 percent of yearly energy", "20 percent of capacity", "seasonal difference of capacity")
totalCapacity	0..1	CEEDS_ActivePower	Information on total capacity in MW.
withConsiderationOfCapacityFromOverlyingGrid	0..1	Boolean	The capacity shown is with (TRUE) or without (FALSE) consideration of available capacity from overlying grid.

LinearCoordinate

Grid Hosting Capacity Document assembly model

Specifies a point on a line segment, including a sequence number and bulge. To visualize the line segment, each linear coordinate is to be connected in the order specified by the sequence attribute. Each linear coordinate must have a sequence value.

Native Members

bulge	0..1	Decimal	Bulge is a factor that describes a curved line segment between two points. bulge=0 is a straight line: bulge=1 is a curve along the edge of a circle. A bulge can be associated only with the second point of a line segment. A negative bulge factor causes the line to bulge to the left as viewed from the first point to the second; a positive bulge causes the line to be curved to the right as viewed from the first point to the second.
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relativeHeight.verticalOffset	0..1	ESMP_Length	Distance above (positive value) or below (negative value) the specified vertical offset reference.
relativeHeight.verticalOffsetReference	0..1	String	The height from which vertical offset is measured. Examples include "Ground Level", "Pole Top", "Tower Top", etc.
sequenceNumber	1..1	Integer	The order in which linear coordinates are to be connected to form the line described. Sequence is required for all linear coordinates.
xPosition	0..1	String	X axis position.
yPosition	0..1	String	Y axis position.
zPosition	0..1	String	(if applicable) Z axis position.

LineElement

Grid Hosting Capacity Document assembly model

A geometric line segment, described by 2 or more coordinate tuples with linear interpolation between them. A line must contain at least 2 linear coordinates.

Native Members

Location

Grid Hosting Capacity Document assembly model

The place, scene, or point of something where someone or something has been, is, and/or will be at a given moment in time. It can be defined with one or more position points (coordinates) in a given coordinate system.

Native Members

mRID	1..1	String	The unique identification of a location. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification, location, etc. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.

MessageDocumentHeader

Grid Hosting Capacity Document assembly model

MessageDocumentHeader is an energy data space specific entity with the creationDateTime of this document.

Native Members

creationDateTime	0..1	ESMP_DateTime	The date and time the MessageDocumenHeader was created.
MetaInformation	[0..1]	MetaInformation	

MetaInformation

Grid Hosting Capacity Document assembly model

MetaInformation is an energy data space specific entity referring to the information concerning the technical and functional context of the MarketDocument.

Native Members

connectionId	0..1	String	connectionId is an energy data space specific attribute referring to the established connection used for the communication of this document.
dataNeedId	0..1	String	dataNeedId is an energy data space specific attribute holding the type of data need requested, like 'last 24 hours at 1 hour interval'.
dataSourceId	0..1	String	dataSourceId is an energy data space specific attribute referring to the datasource from which the requested data is being queried.
defaultValues	0..1	String	defaultValues is a energy data space specific attribute to hold relevant default values. The format to use can freely be chosen based on the implementation. Most commonly you would expect a semi-colon separated name value pair, like: "minPower=value; maxPower=value; avgPower=value".
documentType	0..1	String	documentType is an energy data space specific attribute referring to the type of document, comparable to the MessageTypeList in the CIM part of the message. Note: This is applicable if you have the need to know the content without having to open the complete document. DocumentType might contain messages not (yet) part of the MessageTypeList.
finalCustomerId	0..1	String	The finalCustomerId is an energy data space specific attribute identifying the actor doing the request for data or data sharing.
regionConnector	0..1	String	regionConnector is an energy data space specific attribute that specifies the regional connector being used to retrieve the data requested.

regionCountry	0..1	String	regionCountry is an energy data space specific attribute that specifies the country where the regional connector is located.
requestPermissionId	0..1	String	The requestPermissionId is energy data space specific identification attribute holding an unique identifier issued by the energy data space to allow for tracing of the permission through serveral message exchanges. Note: the requestPermissionId is NOT the identifier issued by the Identity Service Provider as specified in the EU Implementing Act on access to Metering and Consumption Data.
EnvAsset	[0..1]	EnvAsset	

PhysicalAddress
Grid Hosting Capacity Document assembly model

The structured geographic and administrative context of a physical place where an organisation, person, or activity is located.

PhysicalAddress is used for precisely identifying, classifying, and locating entities for regulatory, logistical, or informational purposes.

Native Members (European)

energyIdentCodeEic (European)	0..1	String	The attribute is used for an exchange of the EIC code (Energy identification Code). The length of the string is 16 characters as defined by the EIC code. For details on EIC scheme please refer to ENTSO-E web site.
shortName (European)	0..1	String	The attribute is used for an exchange of a human readable short name with length of the string 12 characters maximum.

Native Members

aliasName	0..1	String	The aliasName is free text human readable name of the object alternative to IdentifiedObject.name. It may be non unique and may not correlate to a naming hierarchy. The attribute aliasName is retained because of backwards compatibility between CIM releases. It is however recommended to replace aliasName with the Name class as aliasName is planned for retirement at a future time.
cityCode	0..1	String	City code.
cityName	0..1	String	Name of the city.
community	1..1	String	Community in which the locality is, and which is in the municipality, city or town.. For example, Anderston, Glasgow City, Scotland or another appropriate fourth-level administrative division
country	0..1	String	Name of the country in which the address is located. It is recommended to use either the 3-letter ISO 3166-1 alpha 3 code such as "SGP," or the full country name such as "Singapore."

county	0..1	String	Name of the second-level Administrative division of a country in which the address is located. The term used for the second-level Administrative division in a county varies from country to country and between first-level Administrative divisions within a country. For example, California has counties, Louisiana has parishes, British Columbia, Canada has regional districts, and Queensland Australia has shires and regions. Counties are usually made up of cities, towns, and rural populations.
description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
fullAddress	1..1	String	A concatenated string specifying the complete address in a single string which may include delimiters.
globalLocationNumber	1..1	String	Global Location Number (GLN, sometimes also referred to as International Location Number or ILN) of the respective organization, person, or place. The GLN is a 13-digit number used to identify parties and physical locations.
identifier	0..1	UUID	A universally unique object identifier. Used to uniquely identify persistent objects between CIM messages.
isicV4	1..1	String	International Standard of Industrial Classification of All Economic Activities (ISIC), Revision 4 code for a particular organization, business person, or place.
isWithinTownLimits	0..1	Boolean	If TRUE, this street is within the legal geographical boundaries of the specified town (default).
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.
postalCode	0..1	String	Postal code for the address.
region	0..1	String	Name of the first-level Administrative division of a country in which the address is located. For example, California, Quebec, and Queensland are states and provinces of the United States of America, Canada, and Australia, respectively.
status	0..1	Status	Current status of the address e.g. pending, approved
streetCode	0..1	String	(if applicable) Utilities often make use of external reference systems, such as those of the town-planner's

			department or surveyor general's mapping system, that allocate global reference codes to streets.
streetName	0..1	String	Name of the street.
streetNumber	0..1	String	Designator of the specific location on the street.
streetPrefix	0..1	String	Prefix to the street name. For example: North, South, East, West.
streetSuffix	0..1	String	Suffix to the street name. For example: North, South, East, West.
suiteNumber	0..1	String	Number of the apartment or suite.

Polygon

Grid Hosting Capacity Document assembly model

One or more self-closing line segments representing a polygonal area. In order to be self-closing, the first and last linear coordinates of each line segment must be identical. A polygon must have one outer boundary and may have one or more inner boundaries. If an inner boundary is specified, the area within the inner boundary is to be excluded from the area within the outer boundary. Each line description within a polygon (both outer and inner boundaries) must contain at least 4 linear coordinates.

Native Members

PositionPoint

Grid Hosting Capacity Document assembly model

Set of spatial coordinates that determine a point, defined in the coordinate system specified in 'Location.CoordinateSystem'. Use a single position point instance to describe a point-oriented location. Use a sequence of position points to describe a line-oriented object (physical location of non-point oriented objects like cables or lines), or area of an object (like a substation or a geographical zone - in this case, have first and last position point with the same values).

Native Members

relativeHeight.verticalOffset	0..1	ESMP_Length	Distance above (positive value) or below (negative value) the specified vertical offset reference.
relativeHeight.verticalOffsetReference	0..1	String	The height from which vertical offset is measured. Examples include "Ground Level", "Pole Top", "Tower Top", etc.
xPosition	0..1	String	X axis position.
yPosition	0..1	String	Y axis position.
zPosition	0..1	String	(if applicable) Z axis position.

ServiceArea

Grid Hosting Capacity Document assembly model

Native Members

identifier	0..1	UUID	A universally unique object identifier. Used to uniquely identify persistent objects between CIM messages.
licenseArea.mRID	0..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
licenseArea.name	0..1	String	The name is any free human readable and possibly non unique text naming the object.
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.
stationName	0..1	String	The human readable name of the station.
UsagePointLocation	[0..1]	UsagePointLocation	Location of this power system resource.
Location	[0..1]	Location	Location of this power system resource.

Site

Area of ground on which work can be executed, organisation can be hosted, and building (including substation and power line) can be located.

Native Members

PhysicalAddress	[1..1]	PhysicalAddress
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Grid Hosting Capacity Document assembly model

UsagePointLocation

Location of an individual usage point.

Native Members

accessMethod	0..1	String	Method for the service person to access this usage point location. For example, a description of where to obtain a key if the facility is unmanned and secured.
description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.

Grid Hosting Capacity Document assembly model

geoInfoReference	0..1	String	(if applicable) Reference to geographical information source, often external to the utility.
mRID	1..1	String	The unique identification of a location. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification, location, etc. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.
type	0..1	String	Classification by utility's corporate standards and practices, relative to the location itself (e.g., geographical, functional accounting, etc., not a given property that happens to exist at that location).
Site	[0..1]	Site	

VoltageLevel
Grid Hosting Capacity Document assembly model

For a production unit or a generating unit, the high voltage level at the network side.
For other network equipment the voltage level.

Native Members

baseVoltage.nominalVoltage	1..1	ESMP_Voltage	The power system resource's base voltage, expressed on a phase-to-phase (line-to-line) basis. Shall be a positive value and not zero. --- The base voltage used for all equipment within the voltage level.
highVoltageLimit	1..1	ESMP_Voltage	The bus bar's high voltage limit
identifier	0..1	UUID	A universally unique object identifier. Used to uniquely identify persistent objects between CIM messages.
lowVoltageLimit	0..1	ESMP_Voltage	The bus bar's low voltage limit
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.

name	0..1	String	The name is any free human readable and possibly non unique text naming the object.
UsagePointLocation	[0..1]	UsagePointLocation	Location of this power system resource.
Location	[0..1]	Location	Location of this power system resource.

Enumerations

[UnitSymbol](#)

[ESMPEnumerations](#)

The coded representation of different units from IEC 61970.

Ampere	The unit of electrical current in the International system of Units (SI) equivalent to one Coulomb per second.
One	A unit for dimensionless quantities, also called quantities of dimension one.
kV	The symbol of kV
MW	The symbol of MW
Percent	A unit of proportion equal to 0.01.
Ohm	The symbol of Ohm Unit
MVAr	The symbol of MVAr
Hertz	HTZ unit as per UN/CEFACT recommendation 20
megavolt-ampere	MVA unit as per UN/CEFACT recommendation 20
degree (unit of angle)	A unit of measurement of angles expressed in a 0 to 360 degree gradient.
grams per kilowatt hour	It represents the mass rate of emissions per unit of work accomplished.

[RoleTypeList](#)

[ESMPEnumerations](#)

Identification of the role played by a party.

Trade responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Consumption responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Combined power exchange (not to be used)	This role is no longer in the ENTSO-E Harmonised Role Model Document.

System operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Imbalance settlement responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Production responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Transmission capacity allocator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Balance responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Metered data aggregator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Billing agent	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Market operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Energy supplier	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Consumer	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Control area operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Control block operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Coordination centre operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Grid access provider	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Grid operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Meter administrator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Party connected to grid	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Producer	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Profile maintenance party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Meter operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Metered data collector	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Metered data responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Metering point administrator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.

Resource Provider	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Scheduling coordinator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Capacity Trader	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Interconnection Trade Responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Nomination Validator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.
Market information aggregator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that collects information from different sources and assembles it to provide a summary of the market.
Information receiver	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party, not necessarily a market participant, which receives information about the market.
Reserve Allocator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that informs the market of reserve requirements, receives tenders against the requirements and in compliance with the prequalification criteria, determines what tenders meet requirements and assigns tenders.
MOL Responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that informs the market of reserve requirements, receives tenders against the requirements and in compliance with the prequalification criteria, determines what tenders meet requirements and assigns tenders.
Capacity Coordinator	A party, acting on behalf of the SOs involved, responsible for establishing a coordinated Offered Capacity and/or NTC and/or ATC between several Scheduling Areas.
Reconciliation Accountable	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that is financially accountable for the reconciled volume of energy products for a profiled local metering point.
Reconciliation Responsible	A party that is responsible for reconciling, within a metering grid area, the volumes used in the imbalance settlement process for profiled metering points and the actual metered quantities.
Data provider	A party that is responsible for providing information to a central authority.
Local Issuing Office (LIO)	A party that is responsible for operating a Local Issuing Office (LIO).
Central Issuing Office (CIO)	A party that is responsible for operating a Central Issuing Office (CIO).
EIC Participant	A party that participates in the EIC environment.
Weather analyser	A party that analyses the current and forecast weather situation and establishes a prognosis of its impact on the renewable energy environment as well as the overall load.
Regional Security Coordinator (RSC)	The RSC as defined in the System Operation guideline.

Energy Service Company (ESCO)	A party offering energy-related services to the Party Connected to Grid, but not directly active in the energy value chain or the physical infrastructure itself. The ESCO may provide insight services as well as energy management services.
Balancing Service Provider	A party with reserve-providing units or reserve-providing groups able to provide balancing services to LFC Operators.
Energy trader	A party that is selling or buying energy.
LFC Operator	A party that is responsible for the Load Frequency Control of its LFC Area or LFC Block.
Transmission System Operator (TSO)	The Transmission System Operator (TSO) is responsible for the transport of electricity on the extra high-voltage and high-voltage interconnected system. This is a market participant and not a role in Harmonised Role Model.
Distribution System Operator (DSO)	Distribution System Operator (DSO) is responsible for transport of electricity on high-voltage (optionally), medium-voltage and low-voltage distribution systems. This is a market participant and not a role in Harmonised Role Model.
Resource Capacity Mechanism Operator	Resource capacity mechanism operator is the party responsible to operate the resource capacity mechanism in a member state. It can either be the TSO or an independent party.
Resource aggregator	A party that aggregates resources for usage by a service provider for energy market services.
Cost sharing calculator	A party responsible for the mapping of the costs between areas and parties.
Settlement responsible	A party responsible for settling the costs.
Coordinated Capacity Calculator	Coordinated Capacity Calculator is the entity or entities with the task of calculating transmission capacity, at regional level or above.
Flexibility service provider	A party that offers flexibility services based on acquired (aggregated) resources.
Data access provider	A party responsible for facilitating access to data.
Metered data administrator	A party responsible for storing and distributing validated measured data.
Permission administrator	A party responsible for administrating a register of data access permissions for a set of metering points, making this information available to final customers and eligible parties in the sector, on request.

<u>CodingSchemeTypeList</u>	<u>ESMPEnumerations</u>
Codification scheme used to identify the coding scheme used for the set of coded values to identify specific objects.	
EIC	The coding scheme is the Energy Identification Coding Scheme (EIC), maintained by ENTSO-E.
Coding scheme which are part of GS1 system	The coding schemes maintained by GS1.
Andorra National coding scheme	The National coding scheme of the country in question.

Albania National coding scheme	The National coding scheme of the country in question.
Armenia National coding scheme	The National coding scheme of the country in question.
Austria National coding scheme	The National coding scheme of the country in question.
Azerbaijan National coding scheme	The National coding scheme of the country in question.
Bosnia and Herzegovina National coding scheme	The National coding scheme of the country in question.
Belgium National coding scheme	The National coding scheme of the country in question.
Bulgaria National coding scheme	The National coding scheme of the country in question.
Switzerland National coding scheme	The National coding scheme of the country in question.
Serbia and Montenegro National coding scheme	The National coding scheme of the country in question.
Czech Republic National coding scheme	The National coding scheme of the country in question.
Germany National coding scheme	The National coding scheme of the country in question.
Denmark National coding scheme	The National coding scheme of the country in question.
Estonia National coding scheme	The National coding scheme of the country in question.
Spain National coding scheme	The National coding scheme of the country in question.
Finland National coding scheme	The National coding scheme of the country in question.
France National coding scheme	The National coding scheme of the country in question.
United Kingdom National coding scheme	The National coding scheme of the country in question.
Georgia National coding scheme	The National coding scheme of the country in question.
Gibraltar National coding scheme	The National coding scheme of the country in question.
Greece National coding scheme	The National coding scheme of the country in question.

Croatia National coding scheme	The National coding scheme of the country in question.
Hungary National coding scheme	The National coding scheme of the country in question.
Ireland National coding scheme	The National coding scheme of the country in question.
Italy National coding scheme	The National coding scheme of the country in question.
Kyrgyzstan National coding scheme	The National coding scheme of the country in question.
Kazakhstan National coding scheme	The National coding scheme of the country in question.
Liechtenstein National coding scheme	The National coding scheme of the country in question.
Lithuania National coding scheme	The National coding scheme of the country in question.
Luxembourg National coding scheme	The National coding scheme of the country in question.
Latvia National coding scheme	The National coding scheme of the country in question.
Morocco National coding scheme	The National coding scheme of the country in question.
Moldavia National coding scheme	The National coding scheme of the country in question.
Macedonia National coding scheme	The National coding scheme of the country in question.
Netherlands National coding scheme	The National coding scheme of the country in question.
Nordic Regional coding scheme	The coding scheme of the Nordic region which covers Denmark, Finland, Norway and Sweden.
Norway National coding scheme	The National coding scheme of the country in question.
Poland National coding scheme	The National coding scheme of the country in question.
Portugal National coding scheme	The National coding scheme of the country in question.
Romania National coding scheme	The National coding scheme of the country in question.
Russian Federation National coding scheme	The National coding scheme of the country in question.

Sweden National coding scheme	The National coding scheme of the country in question.
Slovenia National coding scheme	The National coding scheme of the country in question.
Slovakia National coding scheme	The National coding scheme of the country in question.
Turkey National coding scheme	The National coding scheme of the country in question.
Ukraine National coding scheme	The National coding scheme of the country in question.
CGM	The coding scheme used for Common Grid Model Exchange Standard (CGMES).
Cyprus National coding scheme	The National coding scheme of the country in question.
ISO	The coding scheme for the preceding attribute is a code maintained by International Organization for Standardization (ISO), such as “ISO 3166 <i>English country names and code elements</i> ”.

CustomerConnectionTypeKind

CEEDS_ESMP_CodeLists

Consumption	
Production	
Energy storage	The customer type relevant for the capacity data is Industry customer – energy storage
Hybrid park	

CustomerSubConnectionTypeKind

CEEDS_ESMP_CodeLists

Solar	
Wind	
EV charger	The customer type relevant for the capacity data is EV chargers Light vehicles
Industry	The customer type relevant for the capacity data is Industry customer – high users

CustomerSubSubConnectionTypeKind

CEEDS_ESMP_CodeLists

EV charger light vehicle	The customer type relevant for the capacity data is EV chargers Light vehicles
EV charger heavy duty vehicle	The customer type relevant for the capacity data is EV chargers Heavy Duty

Industry - High usage	The customer type relevant for the capacity data is Industry customer – high users
Industry - Average usage	The customer type relevant for the capacity data is Industry customer – average users

Compound types

Status

[CEEDS_ESMP_DataTypes](#)

Current status information relevant to an entity.

value	0..1	String
dateTime	0..1	DateTime
remark	0..1	String
reason	0..1	String

Datatypes

ESMP_Length

[CEEDS_ESMP_DataTypes](#)

unit	1..1	UnitSymbol	ConstantValue= MTR
value	1..1	Float	

UnitSymbol

[ESMPDataTypes](#)

A code to identify an analog measurement.
A code to uniquely distinguish one occurrence of an entity from another.

In the ESMP context, the code is defined by an emitting company that provides an agreed identification unique within a business context such as capacity auction identification, market agreement identification, etc.

value	1..1	UnitSymbol	Main Core value Space.
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ESMP_DateTime

[ESMPDataTypes](#)

In ESMP, the dateTime shall be expressed in UTC as YYYY-MM-DDThh:mm:ssZ.

value	1..1	DateTime	Main Core value Space.
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ID_String

[ESMPDataTypes](#)

A code to uniquely distinguish one occurrence of an entity from another.

In the ESMP context, the code is defined either by:

- an emitting company that provides an agreed identification unique within a business context such as capacity auction identification, market agreement identification, etc.
- a party (originator of the exchange) that provides a unique identification in the framework of a business exchange such as document identification, time series identification, bid identification, ...

value	1..1	String	Main Core value Space.
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MarketRoleKind_String

ESMPDataTypes

The identification of the role played by a party.

value	1..1	RoleTypeList	Main Core value Space.
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PartyID_String

ESMPDataTypes

The identification of an actor in the energy market.

In the ESMP context, it is an authorized issuing office that provides an agreed identification coding scheme for market participant identification.

codingScheme	1..1	CodingSchemeTypeList	DomainQualification.
value	1..1	String	Main Core value Space.

ESMPVersion_String

ESMPDataTypes

In ESMP, the coded value is restricted to digits.

A code that distinguishes one evolution of an identified object from another. Information about a specific object may be sent several times, each transmission being identified by a different version number.

value	1..1	String	Main Core value Space.
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CEEDS_ActivePower

CEEDS_ESMP_DataTypes

value	1..1	Float	Main core space value. The value is given in decimal format.
unit	1..1	UnitSymbol	ConstantValue= MAW The unit of the value expressed using the UN/CEFACT recommendations 20.

ESMP_Voltage

ESMPDataTypes

The coded identification of a voltage value.

value	1..1	Float	Main Core value Space. The value is expressed as a simple precision and no
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unit	1..1	<u>UnitSymbol</u>	ConstantValue= KVT The unit of the value, the UN/CEFACT recommendation 20 is used as coding scheme of the unit.